

Haroon Muhammed

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EXPERIENCE

Stealth Startup

Oct. 2025 – Present

Robotics Software Engineer

Boston, MA, USA

- Architected a distributed, containerized ROS 2 autonomy stack for a BlueROV2 Heavy integrating Pixhawk with compute split between an NVIDIA Jetson Orin and ground station PC.
- Built a GPU-accelerated perception stack using Nvidia Isaac ROS with cuVSLAM visual-inertial navigation, ESS stereo depth, and EKF sensor fusion for localization.
- Implemented real-time 3D mesh reconstruction using nvblox with TensorRT-optimized segmentation for online semantic filtering, reducing latency via Isaac ROS NITROS.
- Developed a sim-to-real digital twin in Nvidia Isaac Sim with an ArduSub MAVLink backend for closed-loop SITL validation.
- calibrated the intrinsics of an Intel RealSense D455 camera, IMU, and Pixhawk sensors with Kalibr-based extrinsic calibration for sensor fusion.

Boston University (RASTIC)

Jun. 2025 – Aug. 2025

Research Intern

Boston, MA, USA

- Developed an interactive reinforcement learning demonstration in NVIDIA Isaac Sim to visualize training of a Bittle quadruped robot.
- Built a PyQt GUI enabling users to modify learning algorithms, lock joints, and tune hyperparameters during RL training.
- Implemented replay and comparison tools to visualize the impact of learning conditions and physical constraints on training outcomes.

Minus Zero

Jan. 2024 – May 2024

Robotics Intern

Bengaluru, Karnataka, India

- Developed an autonomous vehicle simulation environment in Gazebo and CARLA using ROS2, integrating LiDAR, IMU, ultrasonic sensors, and camera pipelines for perception testing
- Built a containerized ROS 2 perception stack implementing SOTA deep learning models for free-space estimation and lane detection on Indian driving datasets.
- Optimized multi-camera stereo configurations and performed intrinsic/extrinsic calibration using 60° FOV cameras for 360° coverage.
- Configured NVIDIA Jetson Xavier with compatible kernels, drivers, and dependencies for reliable deployment.

LEADERSHIP AND AFFILIATIONS

Programming Lead, Team Dreadnaught Robotics

- Led a team of 20+ students from diverse cultural and ethnic backgrounds, fostering collaboration and innovation.
- Mentored 100+ students in ROS, with several later accepted into competitive robotics research programs.
- Revitalized post-COVID engagement by introducing new projects, increasing team participation by 50%.

Lab Assistant, Boston University (RASTIC)

- Guided graduate students in reinforcement learning projects, assisting with training and simulation setups.
- Mentored 20+ students in ROS, providing hands-on support for robotics projects.
- Configured motion capture systems and robotic vehicles, ensuring seamless integration for research.

PROJECTS

Humanoid Fall Recovery

- Trained a reinforcement learning (PPO) policy in NVIDIA Isaac Lab using RSL-RL to enable a humanoid robot to recover from falls and stand up autonomously in simulation
- Applied curriculum learning and reward shaping to stabilize training and improve recovery success across varied fall scenarios

Behavioral Cloning CARLA

- Developed a data collection pipeline to capture and synchronize LiDAR, IMU, cameras, steering commands, and vehicle telemetry for autonomous driving datasets.
- Implemented DAGGER-based data collection with online dataset balancing across weather conditions, time of day, and driving actions to reduce distribution shift.
- Trained and deployed behavior cloning models for end-to-end autonomous driving control in CARLA environments.

DriveRL-CARLA

- Trained an end-to-end Deep Q-Network (DQN) using an experience replay buffer to improve learning stability and optimize training performance.
- Developed data collection pipelines in CARLA to capture and synchronize IMU, LiDAR, camera, steering angle, and other vehicle telemetry data for training autonomous driving perception and control models.

LLM Powered Robot Arm

- Built an end-to-end ROS 2 manipulation stack integrating GPT-4, perception, and MoveIt, translating natural language commands into collision-aware multi-step robot actions.
- Performed extrinsic calibration between a Sawyer robot arm and an overhead Optitrack camera, for accurate workspace perception and representation

FetchBot: Ball-Catching Robot

- Built a ROS2 robotic system for dynamic ball interception using YOLO-based object detection and trajectory prediction
- Developed PID-based control and localization pipelines using IMU and wheel encoder fusion for autonomous navigation.
- Implemented stereo/monocular camera calibration and physics-based trajectory simulation on Jetson Nano hardware.

Autonomous Underwater Vehicle (TAC Challenge 2023, Norway)

- Developed a digital twin of an autonomous underwater vehicle and simulated environment conditions replicating TAC Challenge 2023
- Deployed an autonomous underwater docking algorithm for aruco marker based docking
- Developed and tuned a PID based control algorithm for a vectored 6dof thruster configured robot

Bridge Game Simulator

- Designed and developed an object-oriented game engine with modular strategy components to support autonomous and human-like player behavior.
- Implemented CI/CD pipelines, automated testing, and version control workflows to improve reliability, scalability, and maintainability.
- Applied object-oriented design principles in Java for backend game logic, while using HTML, CSS, and JavaScript to build the frontend interface.

EDUCATION

Boston University, College of Engineering

Master of Science in Robotics and Autonomous Systems

Aug. 2025
Boston, MA, USA

- GPA: 3.34 / 4.0

Vellore Institute Of Technology, College of Engineering

Btech Computer Science with specialization in AI and Robotics

Aug. 2025
Chennai,TN,India

- GPA: 8.42 / 10.0

SKILLS

Programming : C++, Python, Docker, Git

Robotics & Perception : ROS1, ROS2, MoveIt, Isaac ROS

Simulation & Hardware : Isaac Sim, Isaac Lab, Gazebo, CARLA, Jetson, Raspberry Pi, Intel RealSense, Pixhawk